

Review that CanvasNet Drawing Subsystem Implementation is Correct

2026-10-02

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1 Review that CanvasNet Drawing Subsystem Implementation is Correct

1.1 1. Introduction

1.1.1 1.1 Purpose

This document records the formal review of a set of project files.

1.1.2 1.2 Scope

This is the `CanvasNet-Drawing` ReviewMark review-set: a subsystem-level review proving that the `Drawing` subsystem's top-level design and verification documentation accurately and completely describe what its constituent units provide, and that the subsystem requirements are satisfied — per the hierarchical-scope principle, unit-level implementation detail (source code, tests, and unit design/verification/requirements documents) is explicitly excluded and relied upon only through the subsystem's own description of it.

This is a **re-review** of this review-set, performed from scratch against `feature/pdf-document-support` commit `bc86abc` following a consolidated documentation-sync pass that addressed every finding from the prior review (dated 2026-10-02, fingerprint `5d671b27755d69066b2ce3d64ab02712c034cf141921ecc1e43a44b58017e1ba` , Overall Outcome: Fail). The prior review found that `docs/reqstream/canvas-net/drawing.yaml` had been correctly updated to add the new `TilePaint` unit's requirements, but that the subsystem design document (`docs/design/canvas-net/drawing.md`) and verification design document (`docs/verification/canvas-net/drawing.md`) had not been updated to mention `TilePaint` at all. All three files under review, plus the context files, were re-read in full for this re-review; no conclusions were carried over from the prior run.

1.1.3 1.3 Outcomes

Each check must be recorded with one of the following outcomes:

Outcome	Meaning
Pass	The check was performed and the criterion is satisfied
Fail	The check was performed and the criterion is not satisfied
N/A	The check does not apply; justification is required

1.1.4 1.4 Review Details

Field	Value
Project	CanvasNet
Review ID	CanvasNet-Drawing
Review Title	Review that CanvasNet Drawing Subsystem Implementation is Correct
Fingerprint	5635014228adbd423691e019d839e522c2c86af2b7aae26cb7496a67c4624104
Commit Reviewed	bc86abc1560df73aa141dde47fba8ccbd0f9a2b7 (branch feature/pdf-document-support)
Review Date	2026-10-02
Prior Review	Fail (fingerprint 5d671b27755d69066b2ce3d64ab02712c034cf141921ecc1e43a44b58017e1ba , 2026-10-02) — superseded by this re-review

1.1.5 1.5 Reviewers

Name	Role	Organization	Signature	Date
Formal Review Agent	Automated Formal Reviewer	DEMA Consulting	(agent-generated, no handwritten signature)	2026-10-02

1.1.6 1.6 Files Under Review

File
docs/reqstream/canvas-net/drawing.yaml
docs/design/canvas-net/drawing.md
docs/verification/canvas-net/drawing.md

Context (not under review, for orientation only): docs/design/canvas-net.md , docs/reqstream/canvas-net.yaml , docs/design/introduction.md .

1.2 2. Review Checklist

1.2.1 2.1 Requirements Checks

Applicable: Yes

#	Check	Outcome	Justification
REQ-01	All requirements have a unique identifier	Pass	All IDs in <code>drawing.yaml</code> are unique within the file, including <code>CanvasNet-Drawing-TilePaint</code> and <code>CanvasNet-Drawing-TilePaintValidation</code> .
REQ-02	All requirements are unambiguous	Pass	Each requirement statement is a single, clearly worded sentence.
REQ-03	All requirements are testable	Pass	Each requirement lists concrete <code>tests:</code> entries; all named tests were confirmed to exist in <code>test/DemaConsulting.CanvasNet.Tests/Drawing/</code> (e.g. <code>PathFillerTests.cs</code> , <code>TilePaintEvaluatorTests.cs</code>).
REQ-04	All requirements are consistent	Pass	No contradictions found between <code>TilePaint</code> / <code>TilePaintValidation</code> and the pre-existing <code>GradientPaint</code> / <code>GradientPaintValidation</code> / <code>PathFill</code> / <code>PathStroke</code> requirements.
REQ-05	All requirements are complete (no TBDs)	Pass	No placeholders or TBDs present.
REQ-06	All requirements are verifiable	Pass	Requirements describe objectively observable rendering/validation outcomes.
REQ-07	No compound requirements are present	Pass	Each requirement expresses one coherent testable criterion (validation requirements group related argument-null/range checks sharing one mechanism, consistent with the file's established pattern).
REQ-08	No requirements are missing	Pass	<code>CanvasNet-Drawing-TilePaint</code> and <code>CanvasNet-Drawing-TilePaintValidation</code> are present and are now fully reflected in both the design and verification documents (see DES-01/02/03 and REQDOC-01 below). <code>docs/reqstream/canvas-net.yaml</code> (parent context) links down to both of these requirement IDs (lines ~1198, ~1213), confirming upward traceability is intact. <code>dotnet reqstream --lint --requirements requirements.yaml</code> ran clean

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#	Check	Outcome	Justification
			(exit code 0, no orphaned/unsatisfied requirements reported anywhere in the repository).

1.2.2 2.2 Design Documentation Checks

Applicable: Yes

#	Check	Outcome	Justification
DES-01	Design documentation clearly describes the purpose of the component or feature	Pass	<code>drawing.md</code> 's Purpose section now states the subsystem "currently contains four sibling units" and describes all four (<code>PathFiller</code> , <code>PathStroker</code> , <code>GradientPaint</code> , <code>TilePaint</code>) in the same descriptive style, including <code>TilePaint</code> 's pre-rendered tile <code>Surface</code> , pattern-space transform, and <code>XStep</code> / <code>YStep</code> pitch. The subsystem-overview header paragraph was also updated ("three public entry points... backed by two paint-description sibling units (<code>GradientPaint</code> and <code>TilePaint</code>)").
DES-02	Design documentation covers the necessary implementation details	Pass	The Units section now includes a <code>TilePaint</code> bullet, parallel in structure and detail to the <code>GradientPaint</code> bullet, cross-referencing <code>_TilePaint Unit Design_ (drawing/tile-paint.md)</code> and naming the internal <code>TilePaintEvaluator</code> helper and the non-ownership-of- <code>Surface</code> -lifetime caveat.
DES-03	Design documentation describes how the code is interfaced (APIs, inputs, outputs)	Pass	<code>TilePaint</code> 's public surface (constructor inputs: tile <code>Surface</code> , <code>Transform</code> , <code>XStep</code> / <code>YStep</code>) and its consumption via <code>PathFiller</code> 's tile overload are now described, consistent with the equivalent description already given for <code>GradientPaint</code> . The Dependencies section explicitly states <code>TilePaint</code> depends only on <code>Canvas.Surface</code> and <code>System.Numerics.Matrix3x2</code> , with no dependency on <code>PathFiller</code> / <code>PathStroker</code> / <code>GradientPaint</code> . The Callers section describes <code>TilePaint</code> 's constructor as a public entry point and its consumption by <code>PathFiller</code> 's tile overload, exercised by <code>PathFiller</code> 's own unit tests.
DES-04	Design documentation describes the	Pass	Normal-operation collaboration is clearly described for all four units, including <code>TilePaint</code> 's per-pixel wrapped-around sampling via

#	Check	Outcome	Justification
	expected normal operation		<code>TilePaintEvaluator</code> , analogous to <code>GradientEvaluator</code> .
DES-05	Design documentation describes the expected error handling	N/A	Subsystem-level design intentionally defers error-handling detail to unit design documents per <code>design-documentation.md</code> ; the document does not describe error handling for any of the four units uniformly, which is consistent with the established subsystem-level documentation convention (not a <code>TilePaint</code> -specific gap).

1.2.3 2.3 Technical Documentation Checks

Applicable: No

No general technical documentation files (user guides, API references, README, release notes) are in this review-set's file list; `drawing.md` and `drawing.yaml` are design/requirements artifacts covered by sections 2.2 and 2.1 respectively.

1.2.4 2.4-2.10, 2.12 Code/Test/Security/Readability Checks

Applicable: No

This subsystem-level review-set contains no source code or test files per the hierarchical-scope principle (`.reviewmark.yaml` deliberately limits `CanvasNet-Drawing` 's paths to the subsystem requirements, design, and verification documents only); unit source and tests are reviewed under the separate `CanvasNet-Drawing-TilePaint` , `CanvasNet-Drawing-PathFiller` , `CanvasNet-Drawing-PathStroker` , and `CanvasNet-Drawing-GradientPaint` unit-level review-sets.

1.2.5 2.11 Requirements vs Documentation Checks

Applicable: Yes

Both requirements (`drawing.yaml`) and design/verification documentation (`drawing.md` x2) are in this review-set's file list.

#	Check	Outcome	Justification
REQDOC-01	All reviewed requirements are addressed in the general technical documentation	Pass	<code>drawing.yaml</code> 's <code>CanvasNet-Drawing-TilePaint</code> and <code>CanvasNet-Drawing-TilePaintValidation</code> requirements are now fully addressed in <code>docs/design/canvas-net/drawing.md</code> (Purpose/Units/Dependencies/Callers all describe <code>TilePaint</code>) and in <code>docs/verification/canvas-net/drawing.md</code> (intro, Verification Approach, and Acceptance Criteria all cite <code>_TilePaint Unit Verification Design_</code>). All other requirements in the file (<code>PathFill</code> , <code>PathStroke</code> , <code>GradientPaint</code> , and their validation/no-op variants) remain correctly addressed in both documents.
REQDOC-02	No reviewed requirement is contradicted by the general technical documentation	Pass	No contradictions found between the requirements file and either document.

1.2.6 2.13 Requirements vs Testing Checks

Applicable: Yes

The requirements file links each requirement to named tests; no test source files are in this review-set's scope, so applicability is limited to confirming traceability linkage is present.

#	Check	Outcome	Justification
REQTEST-01	Every requirement under review is covered by at least one test	Pass	<code>CanvasNet-Drawing-TilePaint</code> links to <code>PathFiller_Fill_TilePaint_CheckerboardTile_FillsWithBothTileColors</code> and <code>EvaluateRow_WrapAroundBeyondOneTile_MatchesFirstTileRepetition</code> ; <code>CanvasNet-Drawing-TilePaintValidation</code> links to three named null-argument tests. All named tests were confirmed present in <code>test/DemaConsulting.CanvasNet.Tests/Drawing/PathFillerTests.cs</code> and <code>TilePaintEvaluatorTests.cs</code> . Every other requirement in the file also has a non-empty <code>tests:</code> list. Actual test pass/fail status is unit-level verification evidence, outside this subsystem review-set's scope.
REQTEST-02	Tests verify the behavior described in each requirement	Pass	Test names read as direct paraphrases of their requirement's stated behavior (e.g. <code>TilePaint</code> checkerboard sampling and wrap-around behavior, null-argument rejection).

1.2.7 2.14 Code vs Design Documentation Checks

Applicable: No

No source code files are in this review-set's scope (hierarchical-scope principle); code-to-design consistency for `TilePaint`'s actual implementation is assessed under the `CanvasNet-Drawing-TilePaint` unit-level review-set.

1.3 3. Conclusion

1.3.1 3.1 Summary of Findings

No outstanding findings. All checklist items applicable to this review-set are Pass or appropriately N/A.

#	Observation
Confirmation	<code>docs/reqstream/canvas-net/drawing.yaml</code> remains correctly and completely updated (unchanged conclusion from the prior review): well-formed, testable, non-contradictory <code>CanvasNet-Drawing-TilePaint</code> / <code>CanvasNet-Drawing-TilePaintValidation</code> requirements with correct links, now also mirrored in both design and verification documentation.
Confirmation	<code>docs/design/canvas-net/drawing.md</code> now fully describes four sibling units (<code>PathFiller</code> , <code>PathStroker</code> , <code>GradientPaint</code> , <code>TilePaint</code>) across Purpose, Units, Dependencies, and Callers, resolving the prior review's DES-01/DES-02/DES-03 findings.
Confirmation	<code>docs/verification/canvas-net/drawing.md</code> now cites <code>_TilePaint Unit Verification Design_</code> (<code>drawing/tile-paint.md</code>) alongside the pre-existing three unit verification designs in its intro, Verification Approach, and Acceptance Criteria sections, resolving the prior review's REQDOC-01 finding.
Confirmation	Context file <code>docs/design/canvas-net.md</code> (not under review here, but checked for cross-file consistency/orientation) now correctly states "The <code>Drawing</code> subsystem's four units collaborate as follows" (line 119) — the stale "two units"/"three public entry points" language flagged as an out-of-scope observation in the prior review has also been corrected.
Tooling confirmation	<code>dotnet reqstream --lint --requirements requirements.yaml</code> and <code>dotnet reviewmark --lint</code> both ran clean (exit code 0, no issues reported). <code>npx cspell</code> and <code>npx markdownlint-cli2</code> against all three files under review reported zero issues.

1.3.2 3.2 Overall Outcome

Overall Outcome: Pass

The `Drawing` subsystem's requirements, design, and verification documentation are now mutually consistent and complete with respect to the `TilePaint` unit addition. `docs/reqstream/canvas-net/drawing.yaml` correctly defines `TilePaint` 's requirements; `docs/design/canvas-net/drawing.md` and `docs/verification/canvas-net/drawing.md` both now fully describe `TilePaint` alongside the subsystem's three pre-existing units, with no omissions found in Purpose, Units, Dependencies, Callers, Verification Approach, or Acceptance Criteria. This satisfies the specific compliance question this review-set exists to prove: the `Drawing` subsystem's top-level design and verification documentation accurately and completely describe what its units (including the new `TilePaint` unit) provide, satisfying its own subsystem requirements, excluding unit implementation detail per the hierarchical-scope principle.

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No follow-up actions are required for this review-set.